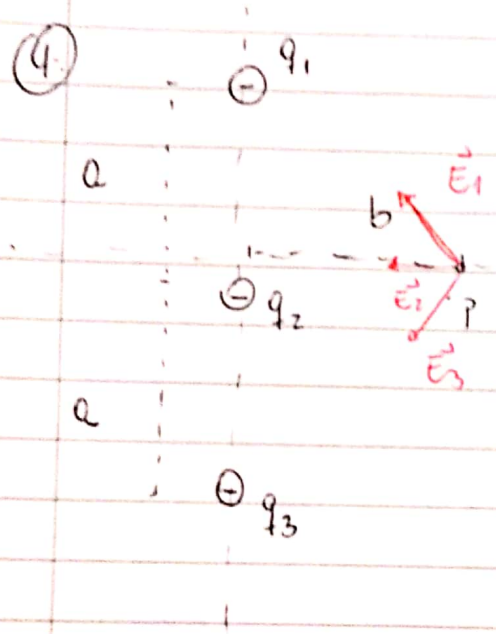
GUÍA 2.

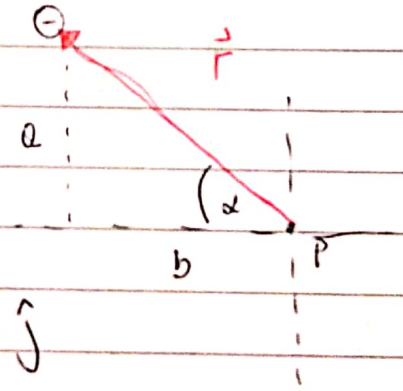
(4)

$a = 0{,}08$ m
$b = 0{,}06$ m



$$\vec{E}_1 = K \frac{q_1}{r_1^3} \vec{r}_1$$

$\vec{r}_1$:



$$\vec{r}_1 = b(-\hat{\imath}) + a\hat{\jmath}$$

$$= 0{,}06(-\hat{\imath}) + 0{,}08\hat{\jmath} \quad [m]$$

$$r_1 = \sqrt{(-0{,}06)^2 + 0{,}08^2}$$

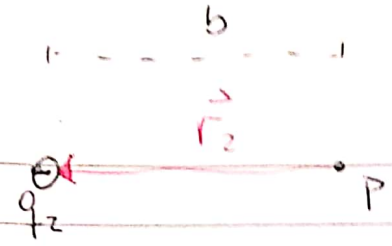
$$= 0{,}1 \text{ m}$$

$$\vec{E}_1 = 9 \cdot 10^9 \frac{5 \cdot 10^{-6}}{(0{,}1)^3} \left(0{,}06(-\hat{\imath}) + 0{,}08\hat{\jmath}\right)$$

$$= 2{,}7 \cdot 10^6 (-\hat{\imath}) + 3{,}6 \cdot 10^6 \hat{\jmath} \quad [N/C]$$

Escaneado con CamScanner

$$\vec{E}_2 = K \frac{q_2}{r_2^3} \vec{r}_2$$

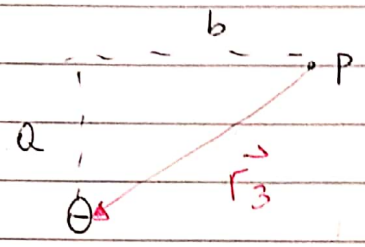


$$\vec{r}_2 = b(-\hat{\imath}) \quad [m]$$
$$= 0{,}06\,(-\hat{\imath})$$

$$\vec{E}_2 = 9 \cdot 10^9 \frac{2 \cdot 10^{-6}}{(0{,}06)^3} 0{,}06(-\hat{\imath})$$

$$= 5 \cdot 10^{+6} (-\hat{\imath}) \quad [N/C]$$

$$\vec{E}_3 = K \frac{q_3}{r_3^3} \vec{r}_3$$



$$\vec{E}_3 = 9 \cdot 10^9 \frac{5 \cdot 10^{-6}}{(0{,}1)^3} [0{,}06(-\hat{\imath}) + 0{,}08(-\hat{\jmath})]$$

$$\vec{r}_3 = b(-\hat{\imath}) + a(-\hat{\jmath})$$
$$= 0{,}06\,(-\hat{\imath}) + 0{,}08\,(-\hat{\jmath})$$

$$= 2{,}7 \cdot 10^{+6} (-\hat{\imath}) + 3{,}6 \cdot 10^{+6} (-\hat{\jmath}) \quad [N/C]$$

$$r_3 = 0{,}1 \quad m$$

$$\Rightarrow \vec{E}_T = \vec{E}_1 + \vec{E}_2 + \vec{E}_3$$

$$= 1{,}04 \cdot 10^7 (-\hat{\imath}) \quad [N/C]$$

$$\Rightarrow \vec{F}_T = q\vec{E}_T$$

$$= (-1 \cdot 10^{-6})\, 1{,}04 \cdot 10^7 (-\hat{\imath})$$

$$= +10{,}4\,\hat{\imath} \quad [N]$$

Escaneado con CamScanner